

Exercise: Day 2

Objective

In this exercise, you will:

1. Retrieve Philippine (PH) varieties from SNP-Seek
2. Create a sample list
3. Subset a VCF file based on selected samples
4. Convert the VCF into a genotype matrix
5. Generate an HDF5 dataset

1. Retrieve Philippine Varieties from SNP-Seek

- Open [SNP-Seek](#)
- Use the **Search by Varieties** function
- Filter or search for varieties from **Philippines (PH)**
- Download the output as CSV file

2. Create a Sample List

- Retrieve the corresponding IRIS IDs for the selected varieties
- Create a text file named **sampleList.txt**
- Each line should contain one sample ID using the format below:
 - IRIS_313-8594
 - IRIS_313-XXXX
 - IRIS_313-XXXX
- Notes:
 - One sample per line
 - No extra spaces

3. Subset the VCF File

- Use:
 - **sampleList.txt** as the list of samples
 - **VCF source:** <https://3kricegenome.s3.dualstack.us-east-1.amazonaws.com/saliksik/riza/base-500k.vcf.gz>
 - **Command:**

```
bcftools view -S <sample_list> <vcf_link> -Oz -o <output_vcf>
```
- Output:
 - Generate a subset VCF file compressed as **.vcf.gz**
 - Output filename: **exerciseD2.vcf.gz**

4. Convert the VCF to a Genotype Matrix

- Use the script **vcftomatrix.sh**
- Expected output:
 - Matrix files inside the output/ directory

5. Generate the HDF5 Dataset

- Use the script **make HDF dataset.sh**
- Expected output:
 - **HDF5 (*.h5)**

Task: HDF5 File Structure Inspection

Inspect the structure of the generated HDF5 file and report its dataspace information.

Ans. _____